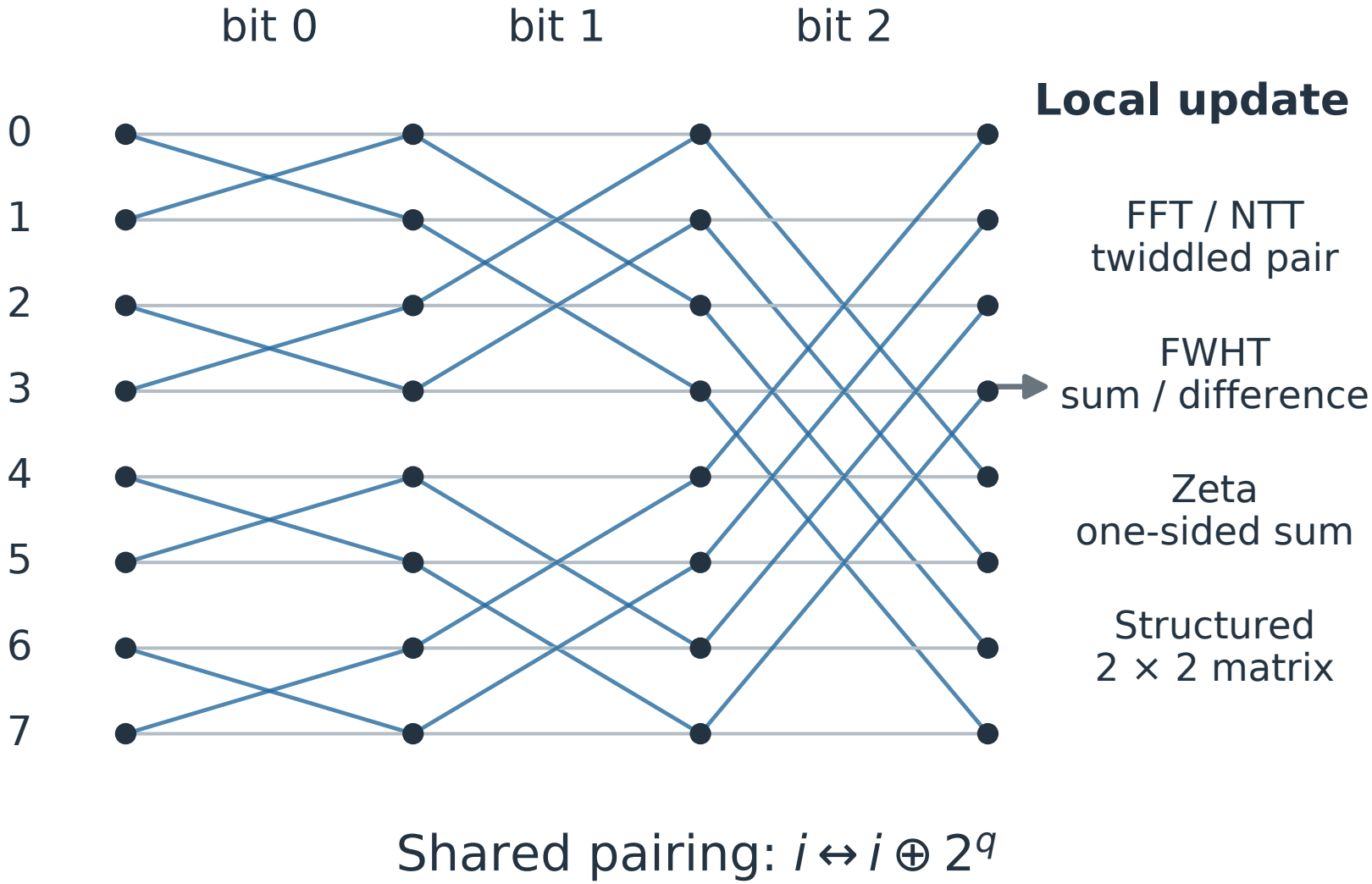


Shared pairing: $i \leftrightarrow i \oplus 2^q$

Arithmetic, order and scaling remain explicit.